

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details : TEAM C

Batch Number : 13

02/02/26

Name: Nishanth Narayana K V

Internship Duration: 8 Weeks

1. Project Title

NeuroFleetX – AI-Driven Urban Mobility Optimization System

2. Project Objective

To build a next-generation AI-driven platform that optimizes urban mobility and fleet operations for rental, transport, and smart city use cases. The system aims to:

- Enable real-time vehicle tracking and intelligent routing across urban zones
- Implement predictive maintenance and dynamic fleet allocation for operators
- Provide customer-centric booking with AI-powered vehicle recommendations
- Support EV fleets with eco-friendly route optimization for maximum battery efficiency
- Reduce traffic congestion — measured by the KPI: average travel time per trip

3. Project description in detail

NeuroFleetX is a next-generation AI-driven platform designed to optimize urban mobility and fleet operations for rental, transport, and smart city use cases. Built entirely as a React single-page application, NeuroFleetX integrates six intelligent modules serving three distinct user roles: Admin, Operator (Fleet Manager), and Commuter (Customer). The system leverages AI logic, data visualization, and geospatial simulation to deliver real-time fleet intelligence.

The system integrates dynamic data rendering, interactive charts, and a simulated map to optimize route plans and monitor vehicle health. It employs advanced dashboards and role-based interfaces for fleet managers and end-users. Designed for scalability, NeuroFleetX can be adapted for electric vehicle (EV) fleets, ride-sharing models, or multi-modal transport networks, offering both operational efficiency and sustainable mobility solutions for modern cities.

Module 1 – Authentication & Role Management

Supports Admin, Fleet Manager, Driver, and Customer roles with secure login and registration. Built using React useState and conditional rendering for role-based dashboard redirection. Each role gets a dedicated dashboard with relevant metric tiles. The login screen features a role dropdown selector, email and password fields, and a gradient Sign In button with the NeuroFleetX branding.

Module 2 – Fleet Inventory & Vehicle Telemetry

Allows adding and updating vehicles, tracking real-time status (Available, In Use, Maintenance), and simulating telemetry data including battery, fuel, and location. Each vehicle is displayed as a grid card with a location pin, status badge (colour-coded), a battery or fuel progress bar computed dynamically from data values, trip count, and Track/Edit action buttons. A filter bar allows instant filtering by vehicle status. An Add New Vehicle form is included at the bottom.

Module 3 – AI Route & Load Optimization Engine

Provides dynamic route suggestions using AI — balancing shortest time, traffic conditions, and energy consumption. The module offers three route options: Fastest Route (minimum travel time, low traffic), Eco Route (optimised for fuel/battery efficiency, ideal for EVs), and Scenic Route (avoids high-density traffic areas). The map screen is rendered using SVG inside JSX, showing colour-coded polyline routes with start and end point markers, a High Traffic Zone overlay, and an AI Insight panel that provides real-time ML-based recommendations.

Module 4 – Predictive Maintenance & Health Analytics

Monitors engine, tyre, fuel, battery, and mileage to predict the next maintenance date and alert the fleet operator before a breakdown occurs. Vehicle health parameters are simulated and evaluated against AI-defined thresholds to trigger severity warnings. The analytics dashboard includes: a Line Chart showing component wear over time (battery, tyres, engine, brakes across 6 months), a Pie Chart showing fleet maintenance status (Healthy, Due Soon, Critical), and an Alert Table listing Vehicle ID, Issue, Severity badge, Recommended Action, and Due Date for each vehicle.

Module 5 – Customer Booking & Smart Recommendations

Customers select vehicle type, date, time, and duration. An AI recommendation engine highlights the best-fit vehicle based on preferences with an AI RECOMMENDED badge. The booking screen includes: All Vehicles / EV Only filter toggle, vehicle cards showing type, seats, star rating, features (Autopilot, Heated Seats, Wi-Fi etc.), and per-hour pricing. A booking form with date picker, time picker, duration slider, and live price calculator allows instant booking. A confirmation screen is shown upon successful booking.

Module 6 – Admin Dashboard & Urban Mobility Insights

Displays real-time fleet intelligence through five KPI cards: Total Fleet (48), Active Routes (17), Trips Today (134), Avg Travel Time (24 min — the primary KPI), and Revenue Today (₹38.4K). Includes a Bar Chart showing hourly

rental activity from 6AM to 10PM, a Trip Density heatmap by zone (Zone A, B, C, D, Depot) using gradient progress bars, a Fleet Health Status Pie Chart, and a Reports & Export panel with CSV and PDF download buttons for Fleet Status, Trip Analytics, Maintenance Log, and Revenue Summary.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project planning, requirement analysis, tech stack selection, React CDN study	Finalised NeuroFleetX concept, studied React CDN basics, designed project structure and module breakdown
Week 2	Build Login and Registration SPA with role selection using React CDN	Completed SPA with Login, Registration, and role selection using useState, conditional rendering, and event handlers
Week 3	Design and build role-based dashboards for Admin, Operator, Commuter	Built three separate dashboards with role-specific views, sidebar navigation, and role badge display
Week 4	Develop Fleet Inventory with vehicle cards and telemetry simulation	Completed fleet grid with status badges, battery/fuel bars, status filter, and Add New Vehicle form
Week 5	Implement AI Route Optimization module with SVG map visualization	Built 3-route system with SVG map, ETA cards, traffic zone overlay, AI insight panel, and route selection state
Week 6	Develop Predictive Maintenance module with health charts and alert table	Completed LineChart for health trends, PieChart for fleet status, and maintenance alert table with severity badges
Week 7	Build Booking module with AI recommendations and confirmation flow	Completed AI recommended vehicle cards, EV filter, booking form with price calculator, and confirmation screen
Week 8	Final testing, Admin Dashboard, recharts integration, GitHub upload, report	Completed KPI dashboard with bar/pie charts, report downloads, GitHub upload via GitHub Desktop, final review

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Requirement analysis done, tech stack finalised, React CDN environment set up	End of Week 1
Prototype/First Draft	Login and Registration SPA with role selection completed using React CDN	End of Week 3
Mid-Term Review	Fleet Inventory, Route Optimization, and Predictive Maintenance modules completed	End of Week 5
Final Submission	All 6 modules complete with Recharts, Admin Dashboard, GitHub upload, and report submitted	End of Week 8
Presentation	Live demo delivered covering all modules, React concepts, chart usage, and KPI outcomes	End of Week 8

5b. Project execution details

The project was executed module by module across 8 weeks using a milestone-driven agile approach.

Module 1

Involved designing the login and registration screens using React CDN inside an HTML file. Implemented role selection during registration (Admin, Operator, Commuter) and role-based dashboard redirection using `useState` and conditional rendering. Later migrated to Create React App with Node.js for better component organisation.

Module 2

Involved creating a vehicle telemetry simulation system. Each vehicle card was rendered dynamically using `.map()` over a mock data array. Battery and fuel progress bars were computed inline from data values (e.g., `width: `${v.battery}%``). Status filter used `useState` to toggle between All, Available, In Use, and Maintenance views.

Module 3

Involved implementing SVG inside JSX to render a simulated route map with three colour-coded polyline paths. Route selection was managed with `useState`, and inline conditional styles were applied to highlight the active route (opacity, stroke-width, border, scale). An AI Insight panel displayed ML-based traffic predictions.

Module 4

Involved installing Recharts via npm (`npm install recharts`) and importing `LineChart`, `PieChart`, `BarChart`, `ResponsiveContainer`, `Tooltip`, `Legend`, `XAxis`, and `YAxis`. Simulated vehicle health parameters over 6 months were plotted using `LineChart`. Alert severity was colour-coded using the reusable `Badge` component with dynamic background and border colours.

Module 5

Involved building a multi-step booking flow managed through multiple `useState` hooks: selected vehicle, date, time, duration (hours), EV filter toggle, and booking confirmation state. Live price was calculated as ``pricePerHr * hours`` and displayed in real time. The AI recommended vehicle was highlighted using a gradient badge positioned absolutely above the card.

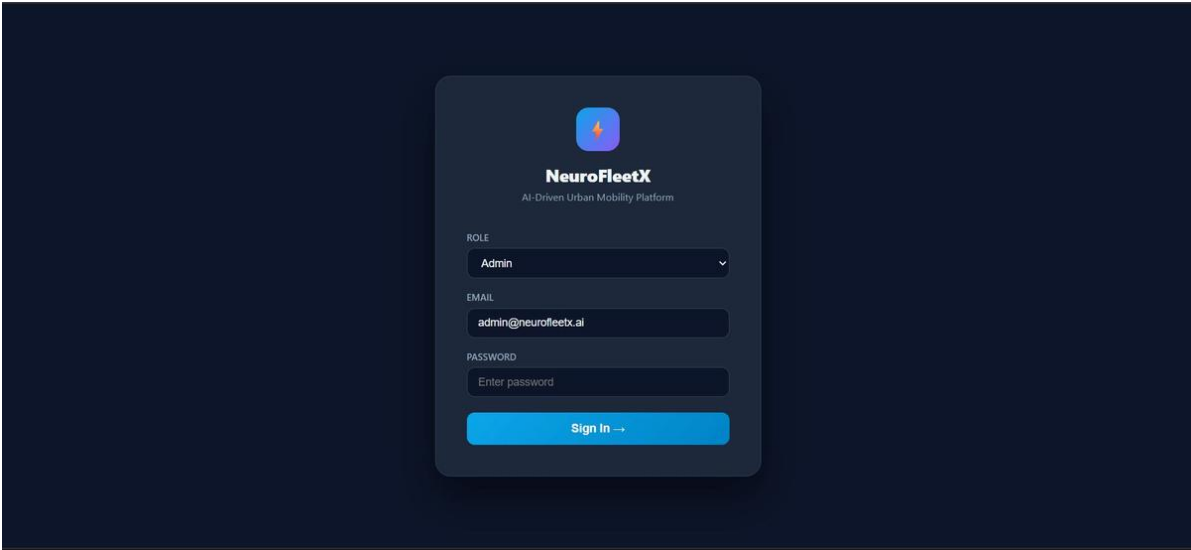
Module 6

Involved creating the Admin Dashboard using React and Recharts. `BarChart` was used for hourly rental activity with rounded bar tops. `PieChart` was used for Fleet Health Status with a donut inner radius. Trip density was visualised using dynamic progress bars computed from percentage values. KPI stat cards were built as a reusable `StatCard` component receiving label, value, sub-text, colour, and icon as props.

6. Snapshots / Screenshots

Login Screen

Dark-themed login page with NeuroFleetX branding, role dropdown selector (Admin / Fleet Manager / Driver / Customer), email and password input fields, and a gradient Sign In button.



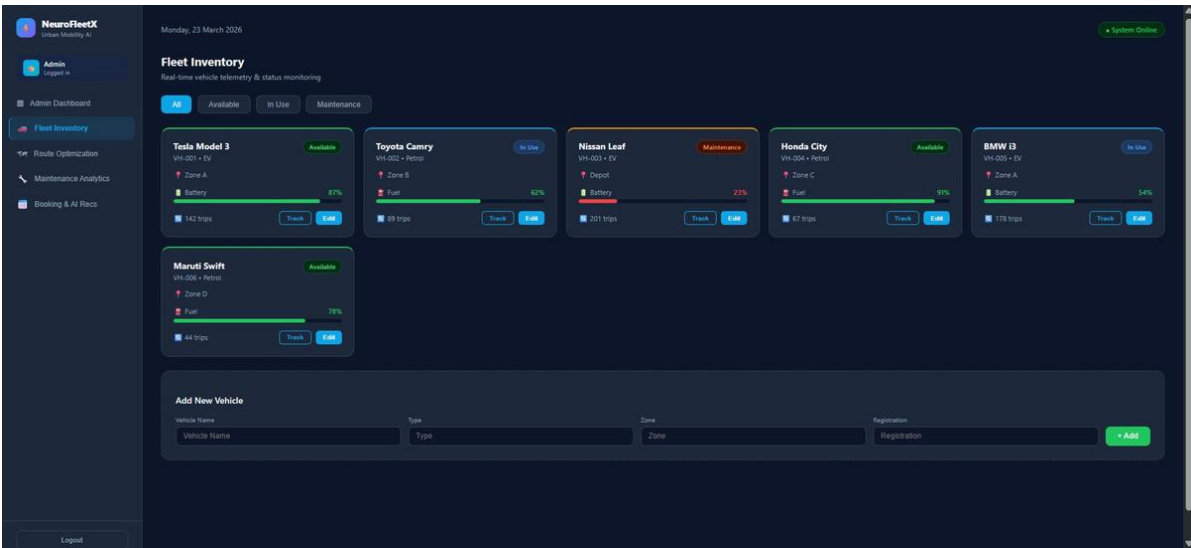
Admin Dashboard

Five KPI cards showing Total Fleet (48), Active Routes (17), Trips Today (134), Avg Travel Time (24 min), and Revenue Today (₹38.4K). Includes hourly rental BarChart, trip density zone bars, Fleet Health PieChart, and report download panel.



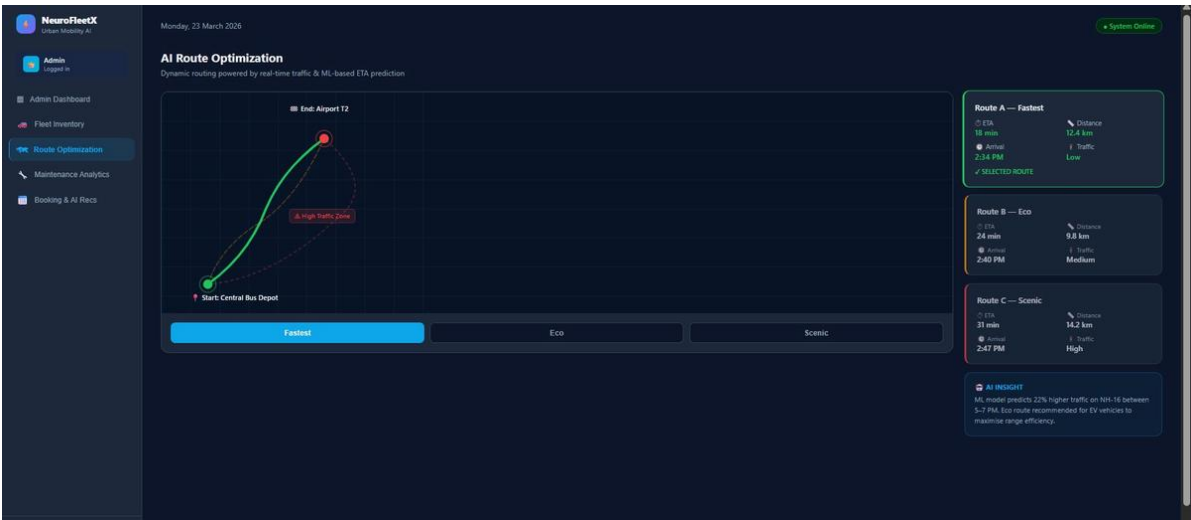
Fleet Inventory

Grid of vehicle cards with colour-coded top borders (green = Available, blue = In Use, orange = Maintenance). Each card shows vehicle name, ID, type, location pin, battery/fuel progress bar, trip count, and Track/Edit buttons. Status filter and Add New Vehicle form included.



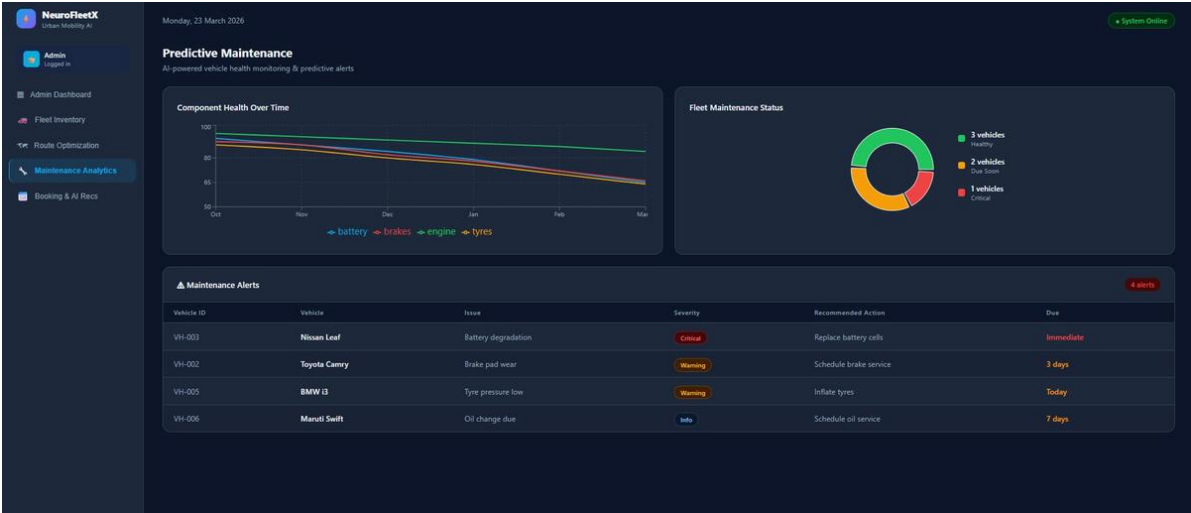
AI Route Optimization

Simulated SVG map with three colour-coded route paths (green = Fastest, yellow dashed = Eco, red dashed = Scenic). Route cards display ETA, distance, arrival time, and traffic level. AI Insight panel provides ML-based real-time recommendations. Fastest/Eco/Scenic toggle buttons at the bottom.



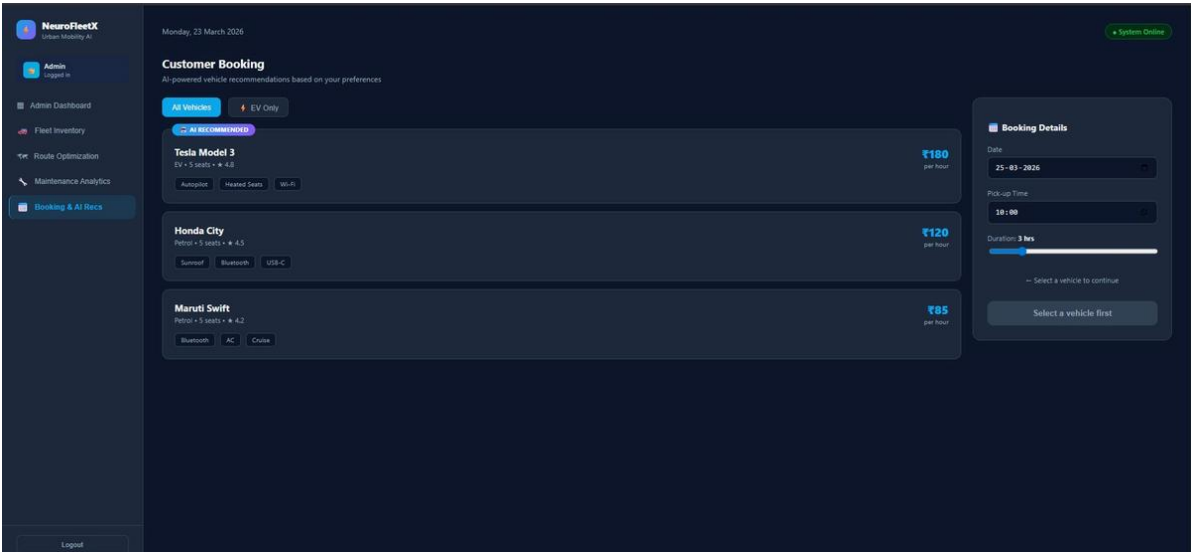
Predictive Maintenance

LineChart showing battery, brakes, engine, and tyre health trends over 6 months (Oct–Mar). Donut PieChart for fleet maintenance status (3 Healthy, 2 Due Soon, 1 Critical). Alert table with Vehicle ID, issue description, severity badge (Critical/Warning/Info), recommended action, and due date.



Customer Booking & AI Recommendations

AI RECOMMENDED badge on Tesla Model 3. All Vehicles / EV Only filter toggle. Vehicle cards with type, seat count, star rating, feature tags, and per-hour pricing. Booking form with date picker, time picker, duration slider, live price calculator ($\text{₹}180 \times 3 \text{ hrs} = \text{₹}540$), and Confirm Booking button.



7. Challenges Faced

- Both App.js and App.jsx existed simultaneously in the project, causing React to always load the old App.js. Resolved by clearing App.js and migrating to a fresh Create React App project with all code in App.js.
- App.jsx was initially saved to the Downloads folder instead of the project src/ directory, causing a Module not found compile error. Resolved by moving the file into the correct src/ folder.
- Recharts was not installed in the initial project, causing unresolved import errors for all chart components. Resolved by running npm install recharts in the VS Code terminal before importing any components.
- Managing multiple useState hooks simultaneously for the booking module (selected vehicle, date, time, hours, EV filter, and booking confirmation) caused unexpected UI behaviour. Resolved by separating each state concern into individual hooks and debugging interactions step by step.
- SVG coordinate calculations for the route map required precise manual positioning of curved path segments and start/end markers. Resolved by calculating coordinates systematically and testing each path incrementally.
- Defining and consistently tracking the KPI (average travel time per trip) across multiple modules required deliberate design decisions at the route, maintenance, and dashboard levels.

8. Learnings & Skills Acquired

Technical:

- React fundamentals: JSX, components, props, useState, conditional rendering, and event handling
- Component-based architecture: reusable Badge, StatCard, and SectionHeader components used across all modules
- Data visualisation using Recharts: BarChart, LineChart, PieChart, ResponsiveContainer, Tooltip, Legend, XAxis, YAxis
- Dynamic rendering: using .map() to generate vehicle cards, alert table rows, nav items, and density bars from data arrays
- SVG inside React JSX: rendering interactive route maps with polyline paths, circles, and zone overlays
- npm package management: installing dependencies, running development servers, and managing Create React App
- Inline dynamic styling: computing styles from data values (battery bar width, badge colour, route highlight opacity)
- Multi-step state management: complex booking flow with multiple independent useState hooks

Domain Knowledge:

- Urban fleet management concepts: vehicle telemetry, route optimization, and predictive maintenance
- AI/ML applications in mobility: recommendation engines, predictive thresholds, and ETA prediction
- KPI-driven development: designing features targeting measurable goals like reducing average travel time
- Role-based access control: building systems where different users see different data and features

9. Testimonials from team

NeuroFleetX was unlike anything I had built before. Starting from basic React CDN usage inside an HTML file and progressively building a full multi-module SPA with AI-driven features, charts, and dynamic data rendering was an incredible journey. Integrating Recharts for data visualisation and SVG for the interactive route map felt like real-world development. Each milestone pushed me to learn something new and apply it immediately to the project. I now feel genuinely confident working on React applications and look forward to building more complex full-stack systems.

10. Conclusion

NeuroFleetX was a comprehensive internship project that gave hands-on exposure to building a full AI-powered platform for urban fleet management. The project successfully delivered all six modules — Authentication, Fleet Inventory, AI Route Optimization, Predictive Maintenance, Customer Booking, and Admin Analytics — using a modern tech stack of React 19, Recharts, SVG in JSX, and npm.

The core goal of the project was to reduce traffic congestion, with average travel time per trip as the primary KPI. Every design and development decision across the 8 weeks was guided by this objective. The route optimization module directly reduces trip time through smart routing, the maintenance module prevents breakdowns that cause delays, and the Admin Dashboard tracks the KPI in real time.

The internship aligned closely with my academic background while introducing me to real-world challenges in React development, AI feature integration, component design, and agile milestone-based delivery. I am confident this experience will be a strong foundation for my career in full-stack web development.

11. Acknowledgements

We extend our heartfelt gratitude to our internship mentor for their constant guidance, technical support, and patience throughout the eight weeks of this program. Their clarity in explaining concepts like KPI definition, component architecture, and AI integration helped us bridge the gap between academic knowledge and real-world development.

We sincerely thank Infosys Springboard for giving us the opportunity to work on a meaningful and challenging project like NeuroFleetX. The structured learning environment, resources, and milestone-based feedback provided to us made it possible to build a platform we are genuinely proud of.

We also thank our fellow interns and team members for their dedication, teamwork, and the many collaborative debugging sessions that turned into some of our best learning moments. This internship has been a defining experience in our journey as software developers.